Form B

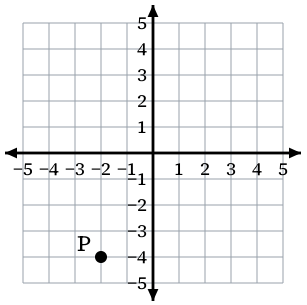
Core Focus on Proportions & Probability

Block 5 Review ~ Direct Variation

Name _____ Period _____ Date _____

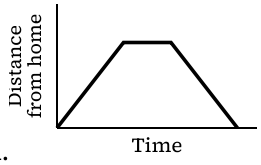
Part I – Selected Response

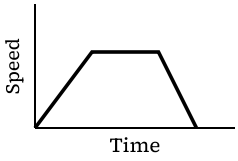
1. Which ordered pair represents the point graphed?

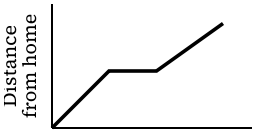


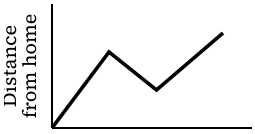
A. (−4, −2)
B. (−2, −4)
C. (2, 4)
D. (−2, 4)

2. Which graph shows a person riding a bike away from home, stopping to rest, and then riding even farther away from home?

A.


B.


C.


D.


3. If the values in the table below model direct variation, what is the rate?

x	0	1	2	3	4
y	0	5	10	15	20

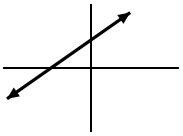
A. 5
B. $\frac{1}{5}$
C. 10
D. It does not model direct variation.

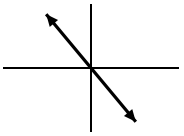
4. Which equation is the direct variation equation for the values in the table below?

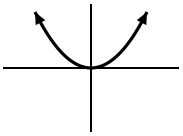
x	0	3	6	9	12
y	0	12	24	36	48

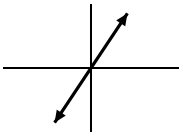
A. $y = 12x$
B. $y = 3x$
C. $y = \frac{1}{4}x$
D. $y = 4x$

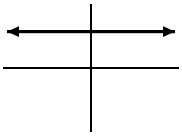
5. Which graphs show direct variation? Circle all that apply.

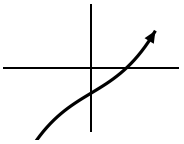
A.


B.


C.


D.


E.


F.


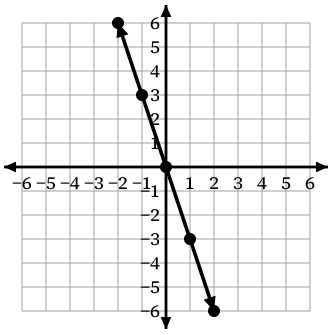
6. Which equations are direct variation equations? Circle all that apply.

A. $y = \frac{x}{4}$
B. $y = \frac{4}{x}$
C. $y = -6x$
D. $y = 2x + 1$
E. $y = 1.5x$
F. $y = 3x^2$

7. What is the slope of the direct variation equation $y = 0.25x$?

A. $\frac{1}{4}$
B. 4
C. 25
D. $\frac{1}{25}$

8. What is the direct variation equation for the line graphed below?



A. $y = 3x$

B. $y = -\frac{1}{3}x$

C. $y = -3x$

D. $y = \frac{1}{3}x$

Form B

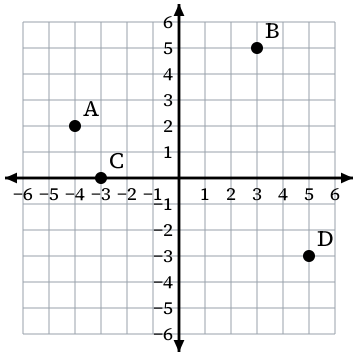
Core Focus on Proportions & Probability

Block 5 Review ~ Direct Variation

Name _____ Period _____ Date _____

Part II – Constructed Response

Write the ordered pair for each point on the coordinate plane to the right.



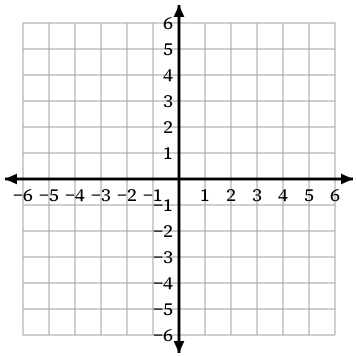
1. A

2. B

3. C

4. D

5. Plot the points (−3, 2), (3, 2), (3, −2) and (−3, −2) and connect them in order. Connect the last point listed to the first point. What shape do the points create?



Shape: _____

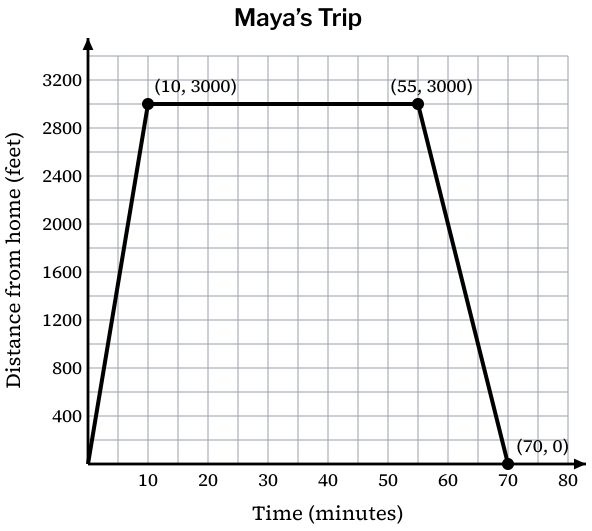
6. Maya walked to the library, studied there, and then walked back home. The graph below shows her distance from home (feet) over time (minutes). For each part, explain how you know your answer is correct.

a. How far away is the library from Maya's house?

b. How long did Maya study at the library?

c. Find Maya's rate of speed in feet per minute for the walk to the library.

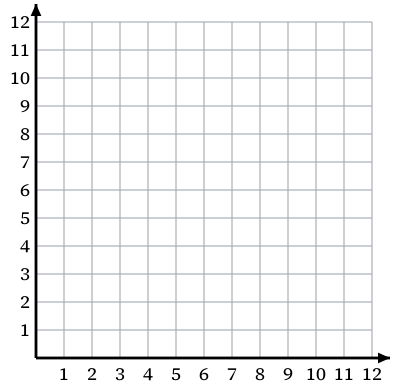
d. Find Maya's rate of speed in feet per minute for the walk home.



Complete each input-output table and graph each function in Quadrant I. What is the slope of each direct variation equation?

7.

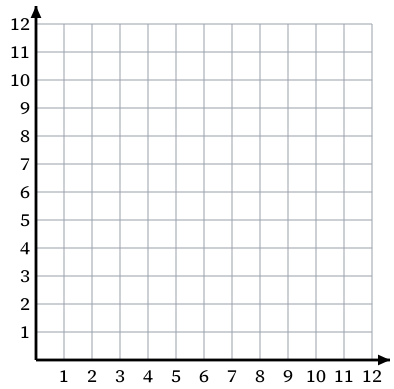
Input x	Function Rule y = 2x	Output y
0		
1		
2		
3		
4		



slope: _____

8.

Input x	Function Rule $y = \frac{1}{2}x$	Output y
0		
2		
4		
6		
8		



slope: _____

Determine whether each table models direct variation. Circle YES or NO. If it does, give the rate. If not, explain why not.

9.

x	y
−2	6
−1	3
0	0
1	−3
2	−6

direct variation? **YES** **NO**

rate/explanation:

10.

x	y
−2	−3
−1	−1
0	1
1	3
2	5

direct variation? **YES** **NO**

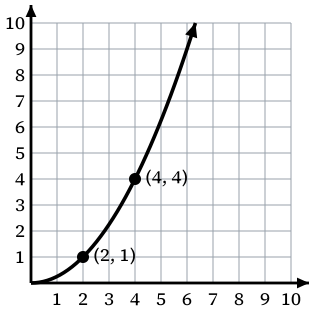
rate/explanation:

Determine whether each equation models direct variation. Circle YES or NO. If it does, give the rate. If not, explain why not.

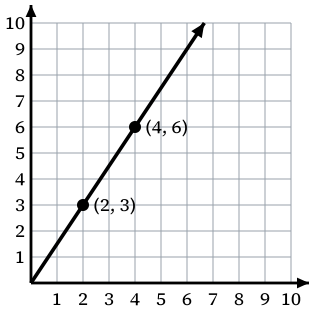
11. y = 0.4x direct variation? **YES** **NO** rate/explanation:

12. y = 5x − 2 direct variation? **YES** **NO** rate/explanation:

13. Ana says Graph B models direct variation and has a rate of $\frac{3}{2}$. Liam says Graph A models direct variation because it passes through the origin. Which person is correct? Explain your reasoning.



Graph A

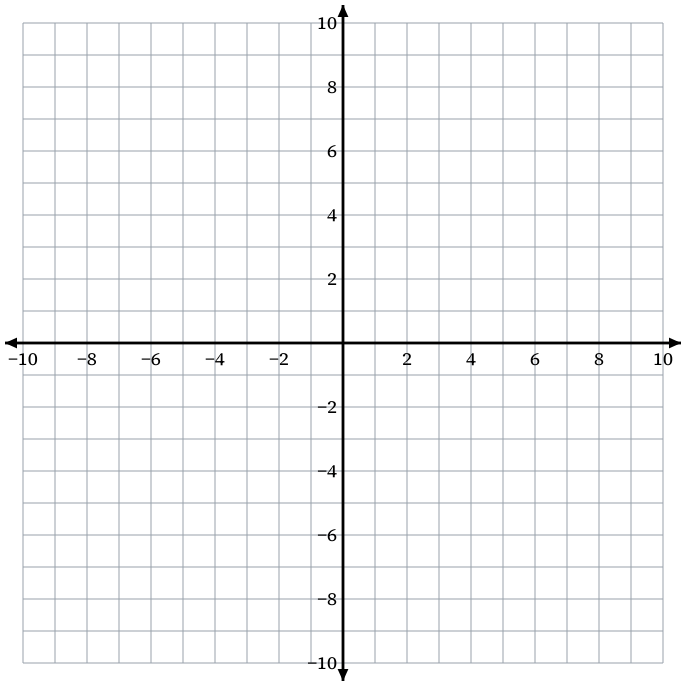


Graph B

14. A printer prints 12 pages every minute. The table below shows the number of pages printed for the number of minutes the printer runs. Write an equation that shows the number of pages, y, printed in x minutes.

x	y
0	0
1	12
2	24
3	36
4	48

15. Graph the direct variation equation $y = -2x$.



Core Focus on Proportions & Probability

Block 5 Review ~ Direct Variation

Name _____ Period _____ Date _____

Part III – Problem Solving

The table below shows the amount of money Jada earned selling bracelets she made.

Bracelets Sold	Money Earned ($)
0	0.00
3	13.50
6	27.00
9	40.50
12	54.00

How many bracelets will Jada need to sell to earn $81.00?